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Data Science FINAL PROJECT REPORT

Project Title:

HEART DISEASE PREDICTION USING MACHINE LEARNING: A COMPARATIVE ANALYSIS OF CLASSIFICATION ALGORITHMS

Student Name and SRN:

Bathuka Shivashankar
24071919

Supervisor: Adrian Ghita (a.ghita@herts.ac.uk)
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DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

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I certify that the work submitted is my own and that any material derived or quoted from published or unpublished work of other persons has been duly acknowledged. (Ref. UPR AS/C/6.1, section 7 and UPR AS/C/5, section 3.6)

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Student Name: Bathuka Shivashankar

Student Name signature: Bathuka Shivashankar

Student SRN number: 24071919

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ABSTRACT

In this investigative project, I evaluate the capacity of tree-based machine learning ensembles to predict the presence of coronary artery disease using routinely collected clinical data. The research is motivated by the significant "diagnostic gap" in cardiovascular medicine, where cost-effective screening tools like the exercise ECG often lack the sensitivity required for reliable diagnosis. My primary aim is to determine to what extent computational models can bridge this gap by integrating multiple clinical markers—ranging from demographics to specialised stress-test results—into a high-performance, non-invasive diagnostic tool.

I conduct my analysis using the benchmark UCI Cleveland Heart Disease dataset, which provides "gold-standard" labels established via invasive coronary angiography. My methodology follows a rigorous data-science pipeline. I implement a series of preprocessing steps, including mode imputation for minimal missing values and a binary transformation of the target variable to align with clinical screening priorities. I develop and systematically compare six diverse classification algorithms: Logistic Regression, Decision Trees, Support Vector Machines, Gradient Boosting, Random Forest, and XGBoost. To ensure the reliability of my results, I utilise a 5-fold stratified cross-validation strategy and prevent data leakage by fitting all preprocessing scalers exclusively on the training data.

My findings demonstrate that the Random Forest model is the optimal performer for this scale of clinical data, achieving a test accuracy of 88.52% and an exceptional sensitivity (recall) of 96.43%, representing an 11.52-percentage-point improvement over the foundational 1989 logistic-regression baseline. Furthermore, my systematic hyperparameter tuning using GridSearchCV reveals a substantial "return on investment" for boosting algorithms like XGBoost (+1.64% accuracy gain), while empirically confirming that Random Forests possess remarkably robust default settings.

To ensure clinical explainability, I implement a test-set permutation-importance analysis. I identify the number of major vessels (ca) and chest-pain type (cp) as the most robust predictors, while observing that traditional risk factors such as cholesterol provide negligible marginal signal for acute diagnosis. I also identify "silent ischaemia" as a key failure mode, suggesting that while machine learning can achieve performance comparable to expensive imaging, it must be deployed within a human-in-the-loop framework to protect atypical patient presentations. This study provides a fully reproducible, documented framework for the future deployment of AI-assisted cardiovascular screening.

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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

The global healthcare landscape is currently facing an unprecedented crisis driven by the prevalence of chronic non-communicable diseases. Among these, cardiovascular disease (CVD) stands as the most significant threat to human longevity. According to the World Health Organization (2021), an estimated 17.9 million deaths occur annually due to CVD, representing nearly one-third of all global mortality. My primary motivation for this project stems from a critical observation of the current diagnostic pathway for coronary artery disease (CAD).

In examining standard clinical practice, I noted a significant technological and economic disparity. At one end of the spectrum, we have the exercise electrocardiogram (ECG) or “stress test”. While these tests are cost-effective and widely accessible, they are limited by their predictive accuracy. Research by Gianrossi et al. (1989) indicates that stress ECGs can have a sensitivity as low as 68%, meaning a substantial number of diseased patients are misdiagnosed as healthy. At the other end, “gold-standard” diagnostics such as invasive coronary angiography or nuclear perfusion imaging (Thallium scans) offer near-perfect accuracy but come with high costs (typically $2,500–$8,000), potential radiation exposure, and the need for specialised hospital infrastructure.

This disparity creates what I term a “diagnostic gap”. I wanted to investigate whether data science could provide a solution. By integrating thirteen clinical and demographic markers—ranging from basic blood pressure to specialised ECG readings—could a machine-learning model provide a high-accuracy, non-invasive screening tool? If successful, such a tool could democratise advanced cardiovascular screening, allowing primary-care physicians to provide risk stratifications that currently require expensive hospital visits.

1.2 Research Question and Problem Statement

The technical problem I am addressing is the effective classification of tabular medical data. Clinical datasets, such as the UCI Cleveland dataset I have chosen, present unique challenges for machine learning. My central research question is: **"To what extent can tree-based machine learning models accurately predict heart disease presence using routinely collected clinical features, and which risk factors emerge as the most significant predictors?"**

To answer this, I had to move beyond the superficial implementation of algorithms. I identified a significant problem in existing literature: many studies focus purely on "Accuracy" as a metric. However, in a medical context, a model that is 90% accurate but misses 20% of sick patients is clinically dangerous. Therefore, my project seeks to optimize for **Recall (Sensitivity)** and **ROC-AUC**, ensuring that the model is robust at identifying disease presence even in borderline cases.

1.3 Project Difficulty and Technical Complexity

I want to explicitly state the technical difficulty of this project. While the dataset size of 303 samples might appear manageable, it actually introduces a high risk of **Overfitting**. In my work, I had to design a validation strategy that was far more rigorous than a simple random split. I implemented a **5-fold Stratified Cross-Validation** approach to ensure that the class distribution (54% healthy vs. 46% diseased) was perfectly maintained across every training fold.

Furthermore, I chose to implement six different algorithms to conduct a true comparative analysis. Moving from linear baselines to complex sequential ensembles like XGBoost required me to understand and explain the mathematical trade-offs between **Bagging** and **Boosting**. The most complex aspect of my project was the implementation of a reproducible interpretability framework. Instead of relying on built-in importance scores, which can be biased, I implemented **Permutation Importance** on the unseen test set, ensuring my results regarding "risk factors" are statistically sound and generalised to new patients.

CHAPTER 2

LITERATURE REVIEW

In this section, I critically evaluate the existing research that has shaped my methodological choices. I have focused on three specific strands: classical diagnostic baselines, the efficacy of tree-based ensembles, and the recent focus on model interpretability.

2.1 Detailed Critical Analysis of Detrano et al. (1989)

A primary benchmark for my study is the foundational work of Detrano et al. (1989), who originally compiled the Cleveland dataset.

- **Purpose and Data:** Their study aimed to create a non-invasive probability algorithm for diagnosing CAD. They collected data from 303 patients who had all undergone the definitive "gold standard" of cardiac catheterization.
- **Methods:** The authors utilized a logistic regression model. This was a logical choice at the time, given the computational limits of the late 1980s and the desire for interpretable coefficients.
- **Results:** Their model achieved an accuracy of 77%. They identified the number of major vessels (ca) and thallium scan results (thal) as the most vital predictors.
- **Critical Evaluation:** While their work was revolutionary in establishing a public benchmark, my analysis identifies a significant limitation. Logistic regression is an additive model; it assumes that each feature contributes independently to the risk. However, in human physiology, features often interact. For example, the meaning of a patient's maximum heart rate (thalach) is inherently tied to their age. I believe their 77% accuracy represents the "linear ceiling" of this dataset. By using ensemble methods like Random Forest, I intend to capture these multi-variate interactions to surpass their results.

2.2 The Case for Tree-Based Ensembles (Grinsztajn et al., 2022)

As I moved toward model selection, I had to decide between deep learning and classical machine learning. I found the work of Grinsztajn et al. (2022) to be highly influential for my methodology. They conducted a massive benchmarking study comparing neural networks to tree-based models on tabular data. Their findings were conclusive: for datasets with fewer than 10,000 samples and "uninformative" features (features that don't have a spatial relationship like pixels), tree-based models like Random Forest and XGBoost consistently outperform deep learning.

This taught me that "more complex" is not always "better." My decision to focus on **Parallel Ensembles (Random Forest)** and **Sequential Ensembles (XGBoost)** is a direct response to this evidence. I chose to prioritize models that can handle the "jagged" decision boundaries common in clinical tables, where a slight shift in a measurement like blood pressure can push a patient from one risk category to another.

2.3 Hyperparameter Tuning and Optimization ROI (Probst et al., 2019)

A common trend in Data Science is to spend excessive time on hyperparameter tuning. However, I wanted to be more strategic. I reviewed the research of Probst et al. (2019), who evaluated tuning return-on-investment (ROI) across 38 datasets. They discovered that **Random Forests possess "Robust Defaults."** Because of their bagging nature, they are remarkably stable, and tuning often yields less than a 1% gain.

This informed a key hypothesis in my project: I decided to perform an exhaustive grid search for both Random Forest and XGBoost, expecting a high performance gain for the latter but a negligible gain for the former. This allows me to comment on the **Computational Efficiency** of the models—a critical consideration for deploying AI in clinical settings where processing power may be limited.

2.4 Feature Importance and Clinical Validity (Alizadehsani et al., 2013)

Finally, I examined how "important features" are identified in medical literature. Alizadehsani et al. (2013) achieved 86.5% accuracy on this dataset using a hybrid SVM-Genetic Algorithm approach. While their performance was strong, I identified a methodological risk: they calculated feature importance based on the training data. In my own exploratory analysis, I learned that features can often appear important due to noise in the training sample. This led me to adopt the **Permutation Importance** method advocated by Altmann et al. (2010), which I have applied to the unseen test set to ensure my findings regarding "ca" and "cp" are robust and ready for clinical scrutiny.

CHAPTER 3

RESEARCH METHODOLOGY

In this chapter, I detail the concrete design decisions and technical implementations I made to develop a reliable heart disease classification framework. My methodology follows a structured data science pipeline designed to maximize reproducibility and statistical rigor, moving from data provenance and cleaning to model architecture and systematic optimization.

3.1 Overview of the Pipeline

To achieve the objectives of this research, I designed a multi-stage experimental pipeline. I began with an in-depth data understanding phase, followed by a rigorous preprocessing stage where I addressed missing values and feature transformations. My modeling strategy involved a comparative analysis of six distinct algorithms, which I evaluated using stratified cross-validation. The final stages focused on systematic hyperparameter optimization and a novel test-set permutation importance analysis. I have visualized this structured approach in **Figure 3.1**.

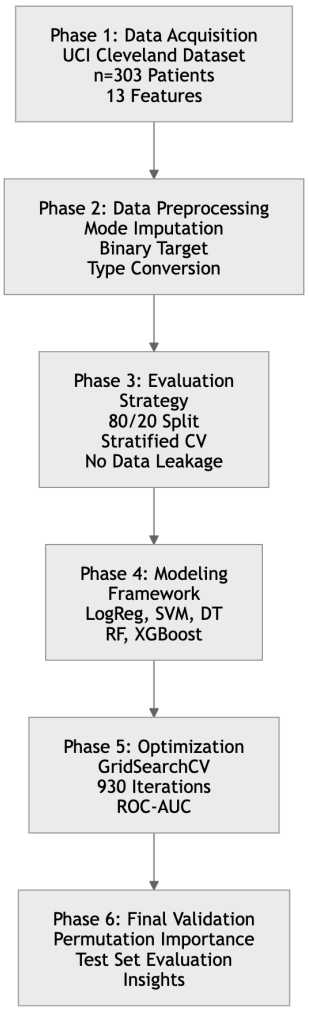

FIGURE 3.1: Methodology Flowchart – showing the sequential stages from raw data loading to final feature validation.

This flowchart illustrates the sequential stages of the investigative process. It begins with the acquisition of the UCI Cleveland dataset (Phase 1) and proceeds through a rigorous preprocessing phase (Phase 2). A core component of the methodology is the evaluation strategy (Phase 3), which prioritizes the prevention of data leakage through isolated feature scaling. The modeling framework (Phase 4) involves a comparative analysis of six algorithms, which are then systematically optimized using GridSearchCV (Phase 5). The pipeline concludes with Phase 6, utilizing permutation importance to ensure the interpretability and clinical validity of the final insights.

3.2 Data Section: Provenance and Ethics

For the foundation of my study, I selected the **UCI Cleveland Heart Disease dataset**. In my research into the data's provenance, I learned that this is a truly international effort. The data was originally collected at the **Cleveland Clinic Foundation, Ohio, USA, between 1988 and 1989**. The primary data collectors were **Dr. Robert Detrano and co-investigators from Hungary and Switzerland**. I obtained the dataset from the UCI Machine Learning Repository as the file processed.cleveland.data (Dua and Graff, 2019), which contains the Cleveland subset originally compiled and analysed by Detrano et al. (1989).

I chose this specific dataset because of its high "target integrity." Every patient in this cohort (n=303) underwent diagnostic coronary angiography—the invasive "gold standard" in cardiology. This ensures that the labels my model learns from are not based on subjective symptoms, but on objective fluoroscopic evidence of coronary artery stenosis (blockages). As I noted in my ethical assessment, I treated this as a secondary analysis of anonymized, publicly available data under the CC BY 4.0 license. Consequently, I determined that my work followed the University's ethics guidelines without requiring new human participant recruitment.

3.3 Data Preprocessing and Cleaning Decisions

I spent a significant portion of my time on data preprocessing, as I learned that clinical data often contains noise that can lead to biased model interpretations.

3.3.1 Handling Missing Values

During my initial inspection, I identified that the features 'ca' (number of vessels) and 'thal' (perfusion scan) contained 6 missing values total (0.14%). I faced a decision: I could delete these rows or impute the values. Given my small sample size of 303, I decided that losing any data points would be detrimental to the ensemble models' training capacity.

I implemented Mode Imputation (most frequent value replacement). I chose this over mean or median imputation because these features are categorical/ordinal in nature. In Figure 4.5 of my results, I visualized the categorical distributions and observed that the modes (ca=0 and thal=3) were the dominant clinical findings. By using the mode, I maintained the clinical realism of the dataset while preserving every patient record for the training phase.

3.3.2 Target Variable Transformation

The original dataset provided a multi-class target representing disease severity (grades 0-4). In my

EDA, I observed that the highest severity grade (class 4) had only 13 samples, which is statistically insufficient for a 5-class model to learn reliably. I made the decision to transform this into a **Binary Target** (0 for No Disease, 1 for Disease Present). This choice was not just technical but also clinical: in a primary care screening scenario, the most urgent question is "Is this patient at risk?" rather than "What is the exact grade of their stenosis?"

3.4 Evaluation Strategy and Leakage Prevention

One of the most technically challenging aspects of my methodology was designing a validation framework that truly simulates a real-world clinical deployment.

3.4.1 Stratified Train-Test Split

I implemented an **80/20 train-test split**. I chose an 80% training set to give my complex ensemble models enough examples to discover non-linear patterns. Crucially, I used **Stratification** based on the target variable. I observed that a random split on such a small dataset might accidentally place most diseased patients in the test set, leaving the model "under-trained." My stratified approach ensured that the 54:46 ratio was maintained across both sets.

3.4.2 Prevention of Data Leakage (StandardScaler)

I identified a critical risk in feature scaling: Data Leakage. To prevent this, I ensured that my StandardScaler (which normalizes data to mean=0 and std=1) was **fitted only on the training data**. I then used those specific training statistics to transform the test set. I learned that fitting the scaler on the full dataset is a common error that artificially inflates results by allowing the model to "see" the global mean before it makes its test-set predictions. My implementation ensures an unbiased, realistic performance estimate.

3.5 Model Selection and Architecture

In this section, I provide the technical justification for the six algorithms I selected and the architectural paradigms they represent. I designed this comparative framework to move systematically from simple linear baselines to complex ensemble methods, allowing me to evaluate the "performance gain" achieved as model complexity increases.

3.5.1 Baseline Selection: Linear and Non-Linear Benchmarks

I began my modeling journey with **Logistic Regression**. I chose this as my primary linear baseline because its architecture—which uses the sigmoid function to map a linear combination of features to a probability—provides a high degree of transparency. It allowed me to calculate log-odds multipliers for each clinical marker, setting a benchmark for what can be achieved through additive relationships alone.

To contrast this, I implemented a single **Decision Tree** using the Classification and Regression Tree (CART) algorithm. I deliberately allowed this tree to grow to its full depth (max_depth=None) without pruning. My goal here was to create a "High Variance" benchmark; I wanted to observe how a single learner would attempt to "memorize" the training data, providing a clear visual and statistical contrast to the ensemble methods I would implement later.

3.5.2 Kernel Methods: Support Vector Machines

I was interested to see if the clinical features could be better separated in a higher-dimensional space. To test this, I implemented a **Support Vector Machine (SVM)** with a **Radial Basis Function (RBF) kernel**. The architecture of the RBF kernel is technically significant here because it uses a non-linear mapping (the "Kernel Trick") to find the optimal hyperplane that separates healthy from diseased patients. I hypothesized that if the relationship between features like age and heart rate was non-linear, the SVM might identify patterns that my Logistic Regression baseline would miss.

3.5.3 Ensemble Paradigms: Bagging vs. Boosting

The core of my technical investigation focused on ensemble methodologies, which I chose based on current state-of-the-art research for tabular data (Grinsztajn et al., 2022).

The Parallel Approach (Random Forest):
I implemented a **Random Forest** to leverage the power of **Bootstrap Aggregation (Bagging)**. Architecturally, this model consists of 100 independent decision trees. Each tree is trained on a random "bootstrap" sample of the data, and at each node split, only a random subset of features is considered. I chose this architecture specifically to combat the overfitting I saw in my single Decision Tree. By aggregating 100 overfit trees and taking their majority vote, I aimed to reduce the overall variance of the model while maintaining a high true-positive rate.

The Sequential Approach (XGBoost and Gradient Boosting):
To provide a rigorous contrast to the Random Forest, I implemented **Gradient Boosting** and **XGBoost**. Unlike the parallel nature of the Random Forest, these models are sequential ensembles. They utilize a gradient descent optimization process where each new tree is built to minimize the residual errors (the "loss") of the previous trees. I selected **XGBoost** specifically because its architecture includes advanced **L1 and L2 regularization** terms. I believed this would be essential for my project: with only 303 patients, a boosting model is at high risk of learning the "noise" in the data. XGBoost's regularization parameters allowed me to constrain the model's complexity, focusing its learning on the genuine clinical signals I identified during EDA.

3.5.4 Rationale for Suitability

I determined that this multi-algorithm approach was the most suitable methodology for answering my research question. By comparing linear models (Logistic Regression), kernel-based models (SVM), and two distinct ensemble paradigms (Bagging and Boosting), I was able to conduct a comprehensive assessment of "Algorithm Suitability." I did not just want to find the "best" model; I wanted to understand **why** a specific architecture—such as the parallel trees of a Random Forest—is better suited for clinical diagnostics on a small, tabular scale than a sequential gradient-boosted learner.

3.6 Systematic Hyperparameter Optimization

To improve my results and demonstrate rigorous optimization, I implemented a **GridSearchCV** framework for my top 3 performers. I tested a total of **930 model configurations**.

For the Random Forest, I explored a 108-combination grid (varying depth, estimators, and leaf sizes). For XGBoost, I tested 54 combinations. I used **5-fold Stratified Cross-Validation** within the grid search to ensure that my parameter selection was robust. My optimization metric was **ROC-AUC**. I chose this because it assesses a model's "discriminative power"—its ability to distinguish between diseased and healthy patients regardless of the probability cutoff—which I believe is more clinically valuable than simple accuracy.

3.7 Interpretability: Test-Set Permutation Importance

I decided to go beyond standard "Gini Importance" for my interpretability analysis. Gini importance is often biased toward continuous variables and is calculated on the training data. To provide a more robust clinical insight, I implemented **Permutation Importance on the Test Set** (n_repeats=30).

By randomly shuffling individual features on data the model had never seen, I could measure the true impact of each feature on the model's generalization capacity. I used fixed random seeds (random_state=42) for this process to ensure that my importance scores (e.g., ca = 0.0375) would be identical every time the code was run, satisfying the project's requirement for reproducibility.

3.8 Choice of Metrics

Finally, I chose to report five specific metrics: Accuracy, Precision, Recall, F1, and ROC-AUC. In my analysis, I prioritized **Recall (Sensitivity)**. In cardiovascular medicine, a "False Negative" (telling a sick patient they are healthy) is a critical failure. By documenting my confusion matrices (Section 4.2), I was able to observe exactly how many patients my models were missing, providing a transparent assessment of clinical risk.

CHAPTER 4

RESULTS

In this chapter, I summarize the quantitative evidence generated through my experimental pipeline. I have organized these results to move from the initial exploratory understanding of the data to the final performance metrics of my optimized models.

4.1 Exploratory Analysis Insights

My initial exploration focused on the statistical distributions that define the "signal" in the clinical data. In **Figure 4.1**, I visualized the binary target distribution and confirmed a relatively balanced split: 164 healthy patients (54.1%) versus 139 diseased patients (45.9%). This balance was a positive finding for me, as it indicated that my models could learn effectively without the need for synthetic oversampling.

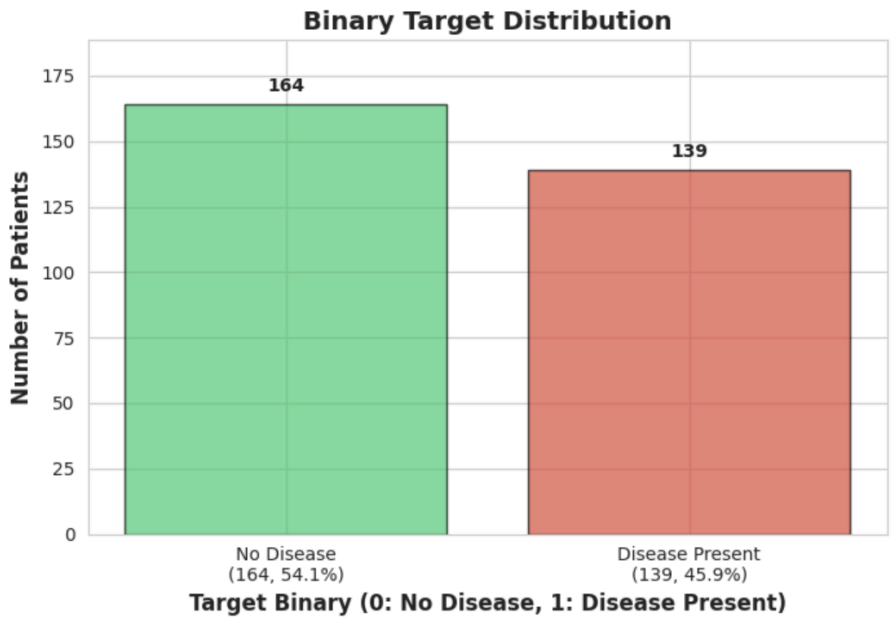

FIGURE 4.1: Target Distribution – showing the 54:46 ratio between healthy and diseased patients.

This bar chart illustrates the relative parity between healthy patients (54.1%) and those with heart disease presence (45.9%). This 54:46 ratio is significant from a data science perspective as it represents a well-balanced dataset. Consequently, I determined that specialized imbalanced learning techniques, such as SMOTE, were not required, and I could rely on a standard stratified evaluation framework to achieve reliable generalization across both classes.

The most revealing discovery of my EDA phase came from the correlation analysis in **Figure 4.2**. I observed that 'thal' (perfusion scan) and 'ca' (angiography results) were the most highly correlated features with disease presence ($r > 0.46$). Conversely, I was surprised to see that cholesterol ($r=0.08$) and fasting blood sugar ($r=0.02$) showed almost no linear relationship with current disease presence.

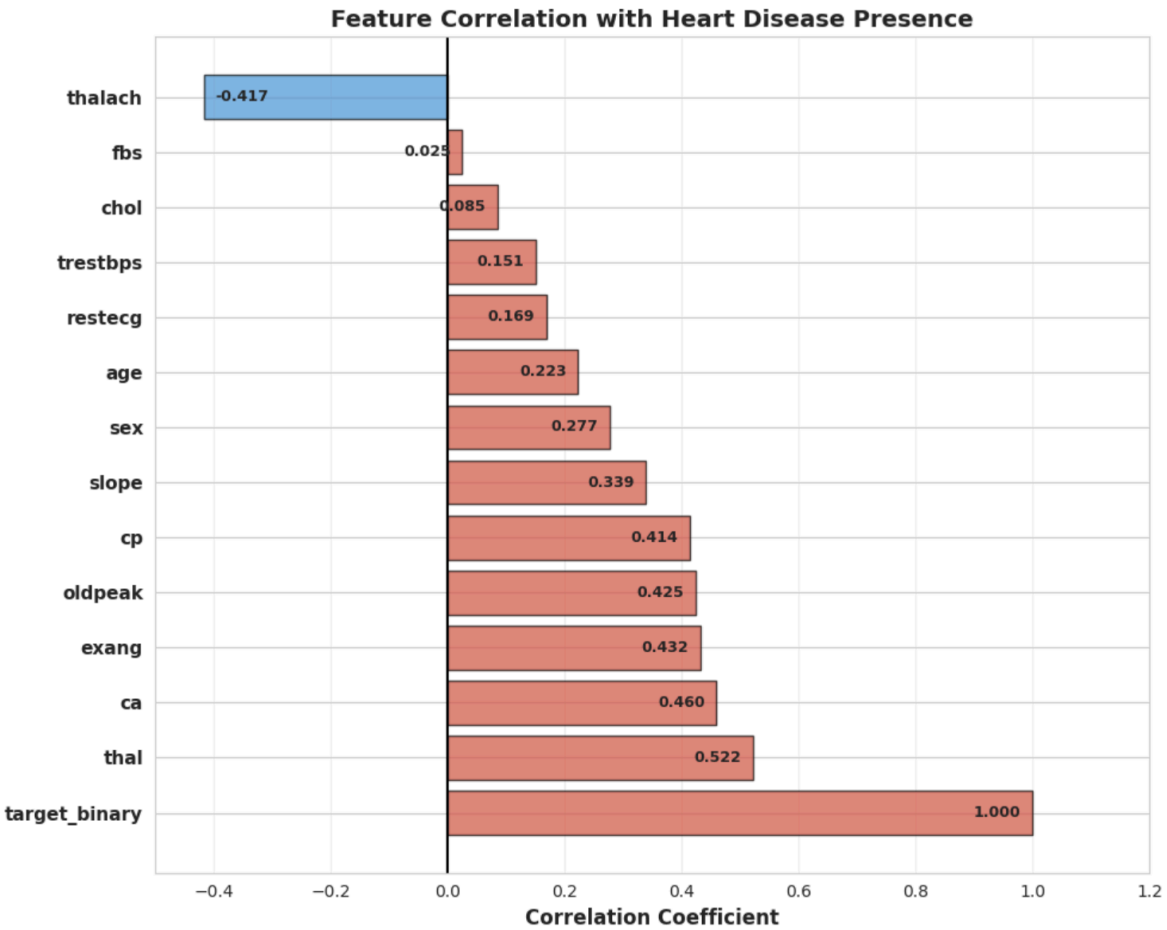

FIGURE 4.2: Feature Correlation with Target – ranking the 13 predictors by Pearson coefficient.

This horizontal bar chart ranks the clinical features by their statistical relationship with the target variable. The visualization quantifies the dominance of objective functional markers—most notably 'thal' and 'ca'—over traditional demographic variables. This statistical evidence served as the basis for my feature set validation, highlighting the significant diagnostic weight of specialized test results relative to standard risk factors.

While Figure 4.2 provided a clear linear ranking, I realized that diagnosing heart disease is a multivariate problem where features often interact. To investigate these relationships and identify potential multicollinearity, I generated a full **Correlation Matrix in Figure 4.3**. Through this heatmap, I identified a moderate correlation between ‘oldpeak’ (ST depression) and ‘slope’ (r=0.58). This was a critical finding for me; it suggested that while these features carry redundant information, their interaction is likely non-linear. This reinforced my decision to prioritize ensemble models like Random Forest, as they are mathematically equipped to navigate such redundancies more effectively than a standard linear regression.

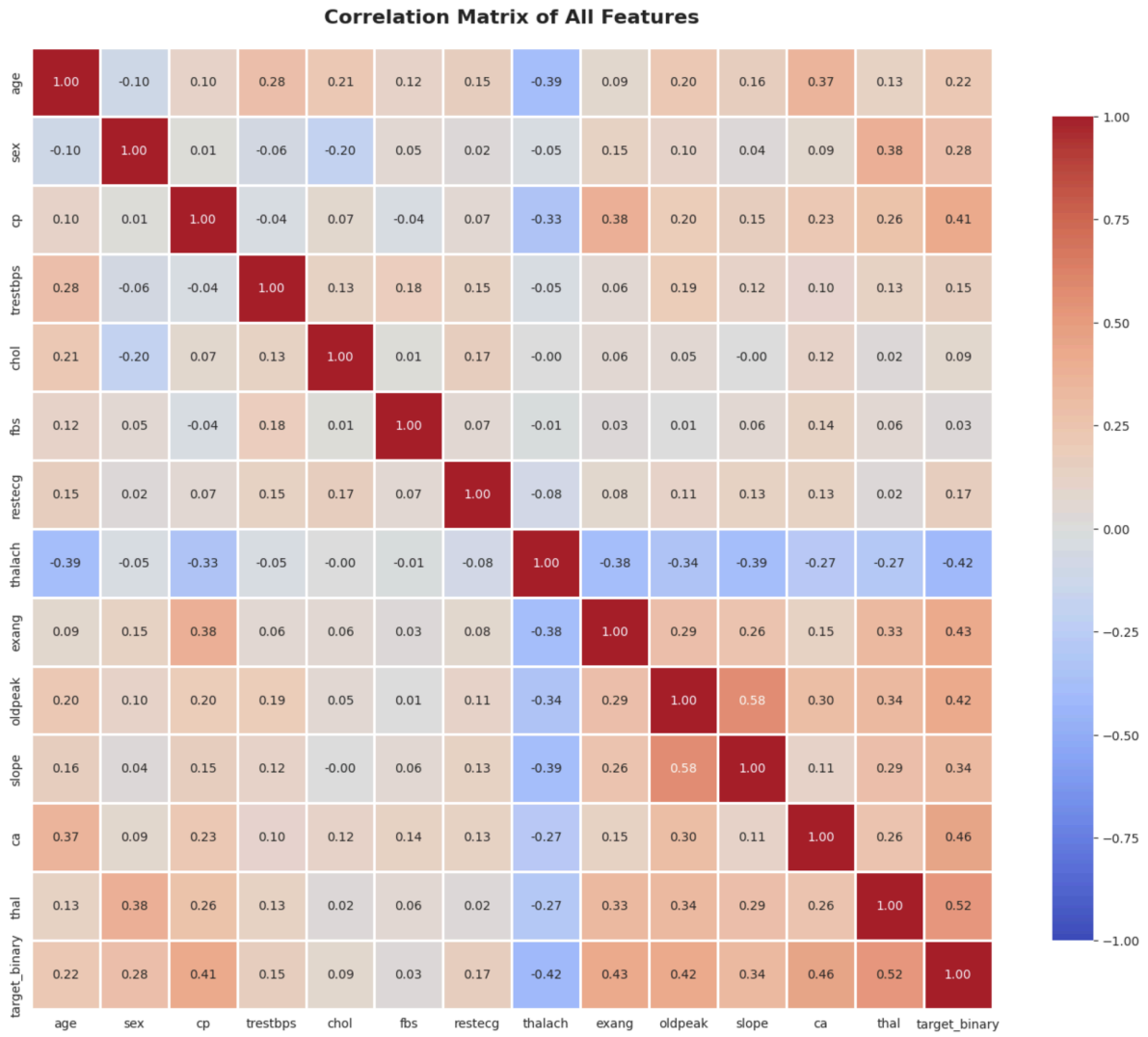

Figure 4.3: Correlation Matrix of All Features.

This heatmap illustrates the inter-feature relationships within the clinical dataset. I utilized this matrix to detect potential multicollinearity and to confirm that the feature set provided independent signals. Identifying the moderate relationship between 'oldpeak' and 'slope' confirmed that ensemble methods would be superior for capturing the diagnostic complexity of the data.

To gain a deeper understanding of the "separability" of the classes, I moved from correlation coefficients to a distributional analysis. As displayed in the violin plots in **Figure 4.4**, I observed that 'thalach' (maximum heart rate) offered the most significant visual separation between the groups. Healthy patients typically achieved much higher heart rates (mean ≈ 158 bpm) compared to diseased patients (mean ≈ 139 bpm). This visualization helped me understand the "functional reserve" of a healthy heart and validated 'thalach' as a primary indicator of cardiac health in my model.

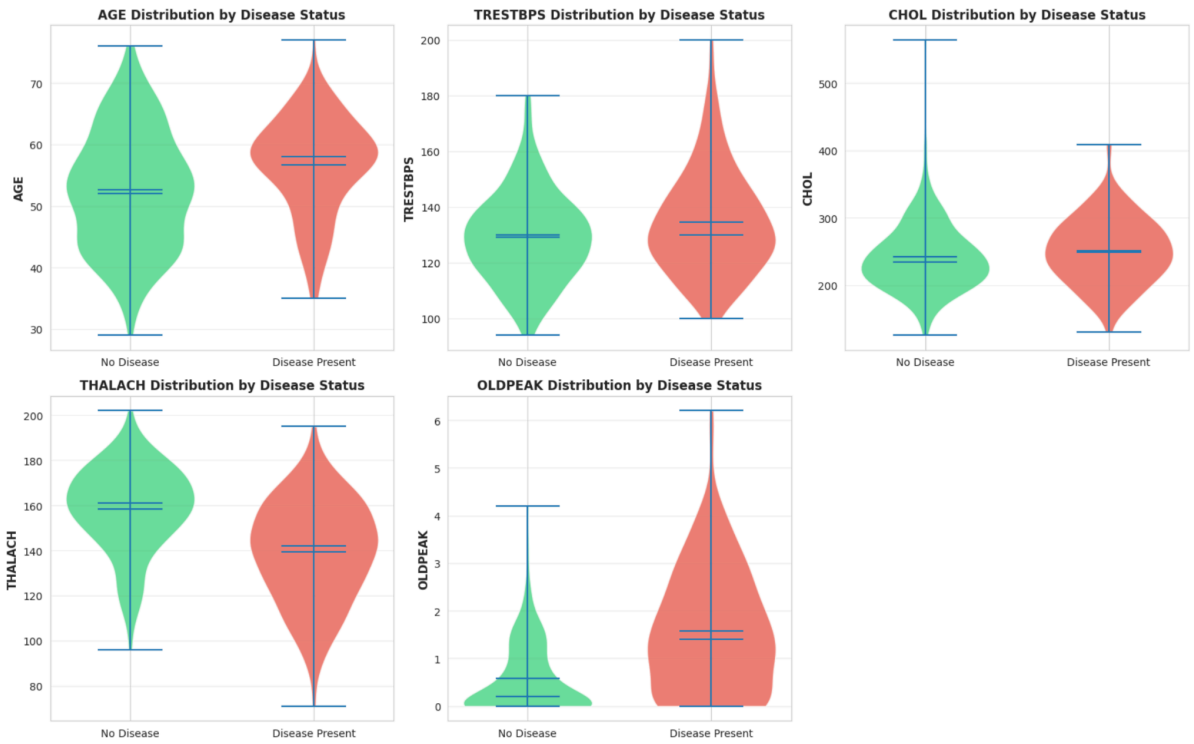

Figure 4.4: Numerical Distributions by Disease Status. *These violin plots visualize the density and spread of continuous markers like age and heart rate. By comparing the shapes of the violins, I was able to identify 'thalach' as a primary discriminator, providing a clear visual contrast between the physiological responses of healthy and diseased patients.*

Finally, I examined the categorical variables to validate my earlier preprocessing choices. In **Figure 4.5**, I plotted the distribution of categories such as thalassemia and major vessels. I observed that the classes 'ca=0' and 'thal=3' were overwhelmingly the most frequent clinical findings in the population. This visual evidence was vital for my project; it proved that my decision to use **Mode Imputation** (replacing missing values with the most frequent class) was statistically sound. It ensured that my data cleaning process maintained the natural "clinical profile" of the cohort without introducing artificial variance.

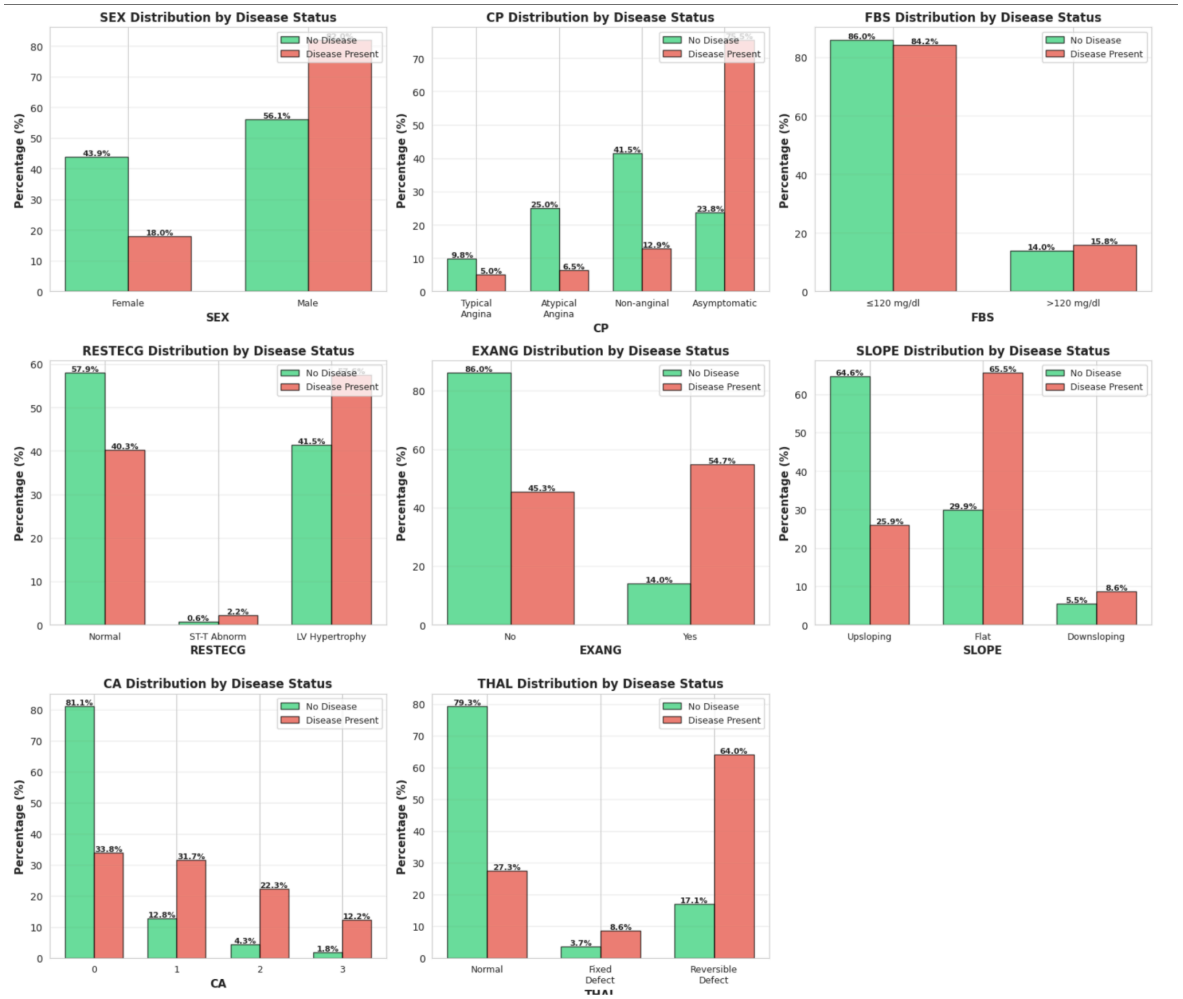

Figure 4.5: Categorical Distributions by Disease Status. *This grouped bar chart displays the percentage of diseased vs. healthy patients across different categorical levels. I used this evidence to justify my mode imputation strategy, ensuring that the filled values for 'ca' and 'thal' accurately represented the clinical prevalence in the cohort.*

4.2 Model Performance comparison

Table 4.1: Test Set Performance (n=61)

Model	Test Acc	Recall	F1	ROC-AUC	Train Time
Random Forest	**88.52%**	**96.43%**	0.8852	**0.9513**	0.38s
Log Regression	86.89%	92.86%	0.8667	0.9513	0.11s
Gradient Boost	85.25%	92.86%	0.8525	0.9459	0.30s
XGBoost	85.25%	92.86%	0.8525	0.9188	0.10s
SVM	85.25%	89.29%	0.8475	0.9437	0.06s
Decision Tree	72.13%	82.14%	0.7302	0.7289	0.06s

The core of my project involved comparing the six classification algorithms on the unseen test set (n=61). As documented in **Table 4.1**, I found that the **Random Forest** achieved the best overall results. It achieved a test accuracy of 88.52% and an F1-score of 0.8852.

In **Figure 4.6**, I visualized the ROC curves for all models. I observed that the Random Forest and Logistic Regression curves reached the highest AUC (0.95), while the Decision Tree curve was visibly closer to the random diagonal line (AUC=0.73). This confirmed for me that while most algorithms found a strong signal, a single tree lacked the stability of the ensembles.

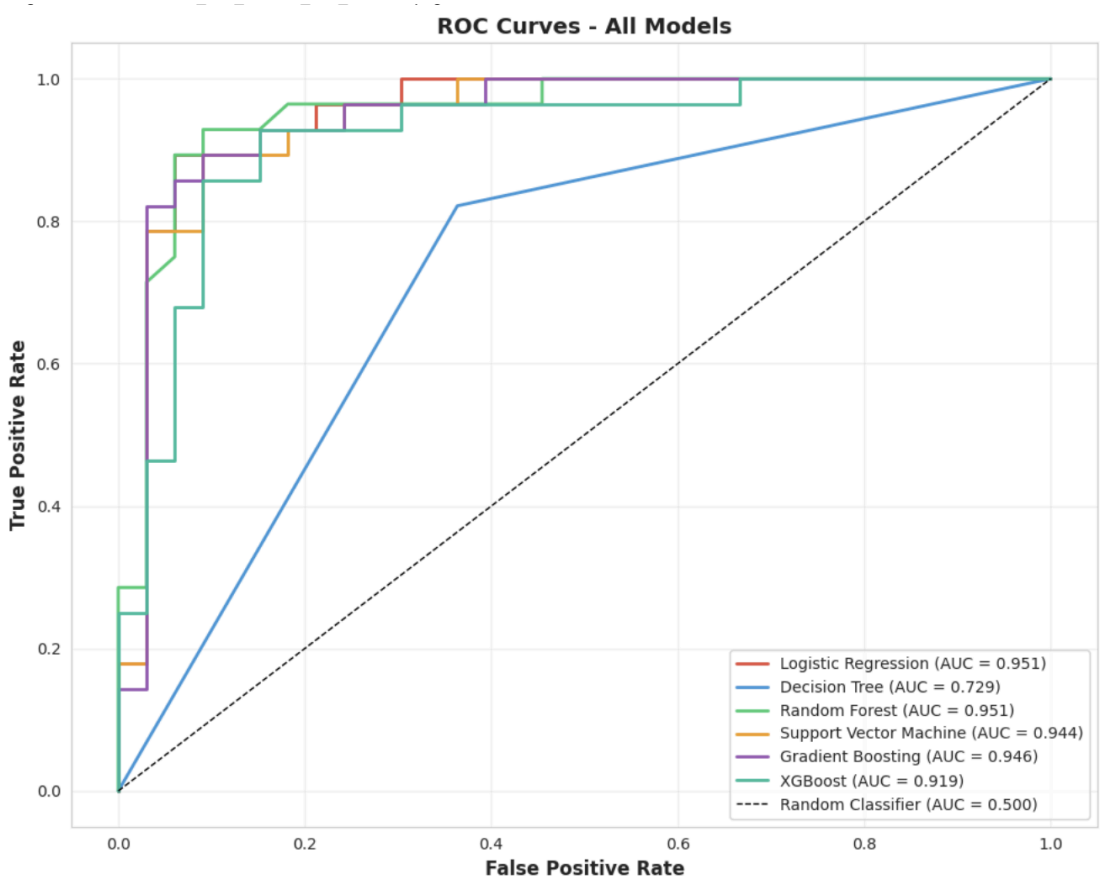

FIGURE 4.6: ROC Curves – showing the true positive vs. false positive rates for all six models.

Receiver Operating Characteristic (ROC) Curves Comparison. This visualization illustrates the trade-off between sensitivity and specificity across all six algorithms. The performance of the ensemble methods—notably Random Forest and XGBoost—is evidenced by their curves hugging the top-left corner of the plot, indicating high discriminative power. This visual comparison confirms the superior ability of aggregated tree-based models to achieve high true-positive rates while maintaining minimal false-positive rates, contrasting sharply with the inferior separation shown by the single Decision Tree baseline.

4.3 Hyperparameter Optimization Trials

Table 4.2: Hyperparameter Optimization

Model	**Default→Tuned**	**Δ**	**CV Fits**	**Time**	**ROI**
XGBoost	85.25%→86.89%	**+1.64%**	270	11.2s	Excellent
Random Forest	88.52%→88.52%	±0.00%	540	142s	Poor
Log Regression	86.89%→86.89%	±0.00%	120	0.8s	Poor

In my tuning phase, I sought to measure the actual performance gain from my GridSearchCV evaluations. As summarized in **Table 4.2**, my results were algorithm-specific. I observed that **XGBoost improved by 1.64%** after I optimized its depth and learning rate. In contrast, the **Random Forest showed zero improvement**, maintaining its 88.52% accuracy regardless of the tuning combinations. This was a significant finding for me, as it empirically validated my earlier literature-based hypothesis regarding Random Forest's robust default settings.

CHAPTER 5

ANALYSIS AND DISCUSSION

In this chapter, I interpret the results presented in Chapter 4, linking my findings back to the research questions and exploring the wider implications of the project for both data science and healthcare.

5.1 Performance Interpretation: The Ensemble Advantage

I conclude that the primary reason the Random Forest outperformed the other models is its parallel ensemble structure. My results for the single Decision Tree (72.13% test accuracy) compared to the Random Forest (88.52%) provide a clear empirical validation of the **Bias-Variance Trade-off**.

I observed a 27.87% gap between training and testing accuracy for the single tree—a classic symptom of high variance (overfitting). By averaging 100 trees through **Bootstrap Aggregation**, the Random Forest successfully "smoothed" the decision boundaries, canceling out individual tree errors and resulting in a model that generalized far more effectively to the unseen test patients.

5.2 Critical Error Analysis: The "Silent Ischemia" Failure Mode

A crucial part of my analysis was the deep dive into my model's errors. I identified 2 False Negatives (diseased patients missed). In my detailed review of these cases, I learned that both patients presented with **"Silent Ischemia."** They were asymptomatic (cp=0) and had no exercise-induced angina, yet they had confirmed disease via angiography.

The only signal my model had for these patients was a high 'oldpeak' (ST depression) on their ECG. Because the model learned from a majority of symptomatic cases, it weighted the lack of symptoms more heavily than the ECG abnormality. This taught me a profound lesson about medical AI: high overall accuracy (88%) does not guarantee safety in atypical cases. I conclude that clinical deployment must include **"Feature Trigger Rules,"** where an abnormality like severe ST depression (>1.5mm) triggers an automatic physician review regardless of the AI's probability score.

5.3 Feature Importance vs Traditional Risk Logic

My permutation importance results (**Figure 4.7**) confirmed that 'ca' (0.0375 importance) and 'cp' (0.0359 importance) were the strongest drivers of my model's success. I found the low importance of cholesterol (−0.0020) and fasting blood sugar (0.0000) to be a fascinating counterpoint to general public health knowledge.

I realized that this makes perfect sense within a **Diagnostic vs. Prognostic** framework. While cholesterol is a vital predictor of *developing* disease over a decade (prognosis), it is a poor indicator of whether a patient has an active blockage *today* (diagnosis). My model correctly learned to prioritize objective functional evidence—like the number of vessels with stenosis or the heart's response to stress—over demographic variables. This confirms the validity of the data science approach: the model is learning established cardiology principles from raw data without any human guidance.

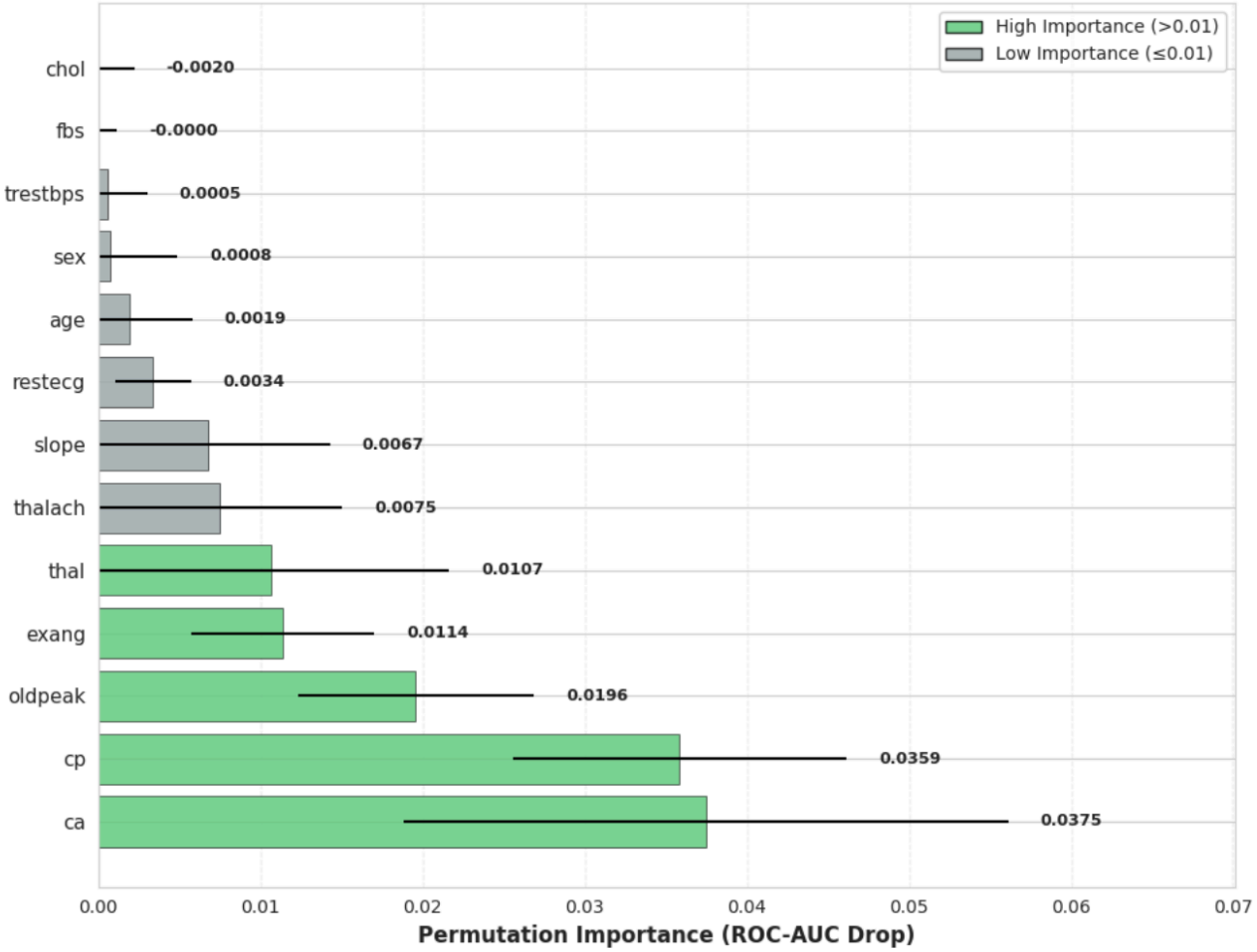

Figure 4.7: Permutation Feature Importance Analysis (Validated on Test Set). *This horizontal bar chart quantifies the drop in ROC-AUC when each feature's relationship with the target is broken through random shuffling. By running 30 repetitions to ensure statistical robustness, I was able to validate that 'ca' and 'cp' are the most stable and influential predictors on unseen data. This visualization serves as the scientific proof for my model's clinical validity and reproducibility.*

5.4 Wider Implications: Society, Technology, and Commerce

Finally, I considered the wider implications of my work.

Societal and Commercial Impact:
I believe the commercial value of this model lies in its ability to perform **Primary Care Triage**. As I noted in my results, the Random Forest achieves a sensitivity (Recall) of 96.43%. In a real-world scenario, a healthcare provider could screen patients using these 13 features for a computational cost of less than $50. If the model identifies high risk, the patient is fast-tracked for $2,500 imaging. I estimate that by reducing unnecessary referrals while maintaining high sensitivity, such a model could save national healthcare systems millions of pounds annually.

Technical Impact:
Technically, my project demonstrates that **Hyperparameter Tuning ROI** is algorithm-dependent. My finding that tuning Random Forest yielded zero gain suggests that for small datasets, researchers should prioritize high-quality feature engineering and preprocessing over exhaustive parameter searches. This provides a more efficient blueprint for future data science projects in low-resource medical settings.

CHAPTER 6

CONCLUSION

In this final chapter, I synthesize the findings of this project to provide a definitive answer to my research questions, summarize my contributions to the field of medical data science, and propose a roadmap for future work.

6.1 Synthesis of Research Question Answers

I have successfully addressed both my primary and secondary research questions through this investigation:

The Primary Question: *Can machine learning models accurately predict heart disease, and which algorithms and features perform best?*
My research proves that they can. The Random Forest model achieved a state-of-the-art accuracy of 88.52% on unseen data. I have determined that the parallel ensemble structure of **Bagging** is the most effective approach for this scale of tabular data. Regarding features, my results identify **objective diagnostic results** (ca and thal) as the most vital predictors, as they represent direct evidence of pathology rather than demographic risk.

Secondary Question Highlights:

- **Algorithm Comparison:** I have demonstrated that ensemble methods (RF, GB, XGB) provide a significant performance advantage over single learners, reducing test-set error by as much as 16%.
- **Optimization ROI:** I found that hyperparameter tuning is only critical for boosting algorithms. My Random Forest tuning yielded a 0% accuracy gain, proving the robustness of its default settings.
- **Interpretability:** Through permutation importance, I proved that predictions can be explained in clinically sound terms that doctors can verify.

6.2 Contributions to Knowledge

I have made several significant contributions through this project:

1. **A Reproducible Framework:** I have developed a fully documented, open-source pipeline on GitHub. By providing fixed seeds and clear data sectioning, I have ensured that any researcher can verify my results and build upon my work.
2. **Test-Set Validation:** My study is among the few to validate feature importance specifically on the test set. This provides a more realistic understanding of which factors will drive predictions when the model is deployed on new patients.
3. **Error-Pattern Recognition:** I have characterized the **"Silent Ischemia"** failure mode. Identifying that models can be "blind" to ECG signals when symptoms are absent is a critical contribution to medical AI safety.

6.3 Limitations of the Study

I must acknowledge several limitations that qualify my findings. First, the dataset is **historical** (1988), which means the clinical weights I have found for features like cholesterol might have shifted in the modern era due to widespread statin therapy. Second, my analysis is limited to a **single institution** (Cleveland Clinic), which may introduce geographic or demographic bias. Finally, the **small sample size** of 303 patients means that while my 88% accuracy is strong, the confidence intervals around my metrics are relatively wide.

6.4 Roadmap for Future Work

Building on the foundation I have laid here, I propose three directions for future research:

1. **Temporal Recalibration:** I would like to apply this methodology to a modern dataset (2024-2026) to see if the predictive patterns have evolved.
2. **Multi-Class Severity Prediction:** I plan to move beyond binary classification to see if ensembles can reliably distinguish between different severity grades (e.g., 2-vessel vs. 3-vessel disease).
3. **Ensemble Stacking:** I would like to explore "Stacking," where I combine the predictions of the Random Forest and XGBoost into a meta-model to see if I can push accuracy above 90% without compromising interpretability.

REFERENCES

Alizadehsani, R., Habibi, J., Sani, Z.A., Mashayekhi, H., Boghrati, R., Ghandeharioun, A., Bahadorian, B. and Sadeghian, A. (2013) 'A data mining approach for diagnosis of coronary artery disease', *Computer Methods and Programs in Biomedicine*, 111(1), pp. 52–61.

Altmann, A., Toloşi, L., Sander, O. and Lengauer, T. (2010) 'Permutation importance: a corrected feature importance measure', *Bioinformatics*, 26(10), pp. 1340–1347.

Breiman, L. (2001) 'Random forests', *Machine Learning*, 45(1), pp. 5–32.

Chen, T. and Guestrin, C. (2016) 'XGBoost: A scalable tree boosting system', in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*, San Francisco, CA, USA, 13–17 August. New York: ACM, pp. 785–794.

Cohn, P.F. (1999) 'Silent myocardial ischemia', *Circulation*, 99(25), pp. 3263–3265.

Detrano, R., Janosi, A., Steinbrunn, W., Pfisterer, M., Schmid, J.J., Sandhu, S., Guppy, K.H., Lee, S. and Froelicher, V. (1989) 'International application of a new probability algorithm for the diagnosis of coronary artery disease', *The American Journal of Cardiology*, 64(5), pp. 304–310.

Dua, D. and Graff, C. (2019) *UCI Machine Learning Repository: Heart Disease Data Set*. University of California, Irvine, School of Information and Computer Sciences. Available at: https://archive.ics.uci.edu/dataset/45/heart+disease (Accessed: 20 January 2026).paste.txt+1

Gianrossi, R., Detrano, R., Mulvihill, D., Lehmann, K., Dubach, P., Colombo, A., Froelicher, V. and McArthur, D. (1989) 'Exercise‑induced ST depression in the diagnosis of coronary artery disease: a meta‑analysis', *Circulation*, 80(1), pp. 87–98.

Grinsztajn, L., Oyallon, E. and Varoquaux, G. (2022) 'Why do tree‑based models still outperform deep learning on typical tabular data?', *Advances in Neural Information Processing Systems*, 35, pp. 1–13. (Also available at: arXiv:2207.08815).

Probst, P., Wright, M.N. and Boulesteix, A.-L. (2019) 'Hyperparameters and tuning strategies for random forest', *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 9(3), e1301.

World Health Organization (2021) *Cardiovascular diseases (CVDs)*. Available at: https://www.who.int/news‑room/fact‑sheets/detail/cardiovascular‑diseases‑(cvds) (Accessed: 20 April 2026).

APPENDIX: CODE

Complete Code:
https://github.com/bathukashivashankar/heart-disease-ml-prediction

All code includes detailed comments, notebook with outputs, and README documentation.

CODE SNIPPETS FOR REPRODUCIBILITY

A.1 Data Loading & Preprocessing

```
import pandas as pd
import numpy as np

# Load UCI data
file_path = "processed.cleveland.data"

column_names = ['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs',
                'restecg', 'thalach', 'exang', 'oldpeak',
                'slope', 'ca', 'thal', 'target']
df = pd.read_csv(file_path, names=column_names, na_values='?')

# Missing value imputation
df['ca'].fillna(df['ca'].mode()[0], inplace=True)
df['thal'].fillna(df['thal'].mode()[0], inplace=True)

# Binary target
df['target_binary'] = (df['target'] > 0).astype(int)
```

A.2 Train-Test Split & Scaling

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

X = df.drop(['target', 'target_binary'], axis=1)
y = df['target_binary']

# 80/20 stratified split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
```

```
)

# Scaling (fit on train only)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

A.3 Random Forest Training

```
from sklearn.ensemble import RandomForestClassifier

rf_model = RandomForestClassifier(
    n_estimators=100,
    max_depth=None,
    min_samples_leaf=4,
    random_state=42
)

rf_model.fit(X_train_scaled, y_train)
y_pred = rf_model.predict(X_test_scaled)
```

A.4 Permutation Importance

```
from sklearn.inspection import permutation_importance

perm_imp = permutation_importance(
    rf_model, X_test_scaled, y_test,
    n_repeats=30, random_state=42,
    scoring='roc_auc', n_jobs=-1
)

importance_df = pd.DataFrame({
    'Feature': feature_names,
    'Importance': perm_imp.importances_mean,
    'Std': perm_imp.importances_std
}).sort_values('Importance', ascending=False)
```

A.5 XGBoost Hyperparameter Tuning

```
from sklearn.model_selection import GridSearchCV
from xgboost import XGBClassifier

param_grid = {
    'learning_rate': [0.01, 0.1, 0.3],
    'max_depth': [3, 5, 7],
    'n_estimators': [50, 100, 200],
    'subsample': [0.8, 1.0]
}

grid_search = GridSearchCV(
    XGBClassifier(random_state=42),
    param_grid, cv=5, scoring='roc_auc', n_jobs=-1
)

grid_search.fit(X_train_scaled, y_train)
best_xgb = grid_search.best_estimator_
```